



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 13/09/2016*

EXERCISE

We observe N samples of a realization of the discrete-time random process $X[n] = A \cos(2\pi F_0 n + \theta) + W[n]$, for $n=0,1,\dots, N-1$. A is the amplitude and θ the initial phase of the signal of interest (SOI), they are modelled as unknown deterministic parameters. The digital frequency F_0 is assumed to be a priori known ($0 < F_0 < 1/2$). $\{W[n]; n=0,1,\dots, N-1\}$ are independent random variables modeling additive noise; each noise sample is Gaussian distributed, with zero mean and variance σ_w^2 .

- (1) Derive the joint Maximum Likelihood (ML) estimator of A and θ given the observed data vector $\mathbf{X} \triangleq [X[0] X[1] \dots X[N-1]]^T$.
- (2) Derive the Cramér-Rao lower bounds (CRBs) on the estimate of A and θ and express them as a function of the signal to noise ratio (SNR): $\gamma = A^2/(2\sigma_w^2)$.
- (3) What are the properties of the two ML estimates?